

Virgo 120M Fine-Tuning Dataset Roadmap: Current State to 300K Instruction Pairs

1. Current Dataset (Already Available)

Category	Current Size	Status
Markdown Generation	~10k	Strong
Structured Documents	Included	Strong
Short Response Control	~40k+	Strong
Identity / Behavior	~5k	Good
Technical Explanations	Included	Good
Current Total	~65k	Existing

2. Additional Data Required

Category	Add Samples	Purpose
Natural Conversation	80,000	Human-like assistant behavior
General Instruction Following	60,000	Explain, summarize, create, rewrite
Coding & Software Engineering	25,000	Programming and debugging
Science & Mathematics	20,000	Technical knowledge
Professional Writing	20,000	Emails, letters, reports
Reasoning & Planning	15,000	Problem solving
Formatting Control	10,000	JSON, YAML, structured outputs
Identity & Safety	10,000	Behavior consistency and safe responses
Total Addition	240,000	New samples required

3. Final Dataset Composition

Source	Samples
Existing Dataset	65,000
Chat Dataset	80,000
Instruction Dataset	60,000
Coding	25,000
Science/Math	20,000
Professional Writing	20,000
Reasoning	15,000
Formatting	10,000
Identity/Safety	10,000

Source	Samples
Final Total	~300,000

4. Final Category Distribution

Category	Samples	Percentage
Conversation	80k	27%
General Instruction	60k	20%
Markdown/Documents	25k	8%
Coding	25k	8%
Science/Math	20k	7%
Professional Writing	20k	7%
Reasoning	15k	5%
Formatting	10k	3%
Identity	5k	2%
Safety/Evaluation	10k	3%

5. Creation Priority Order

Phase	Tasks
Phase 1	Create 80k chat + 60k instruction dataset
Phase 2	Create coding, science, professional writing datasets
Phase 3	Add reasoning, formatting, safety datasets
Final Goal	Train Virgo 120M on ~300k high-quality pairs

Recommendation: Do not only increase dataset size. Maintain category balance and prioritize high-quality conversation and instruction samples because a 120M parameter model benefits more from diversity and clean supervision.